1.

a) Index

500H

Base = B56700 - Start

End = Base + limit

= B56700 + 0129

= B56829H

∴ B123AB H does not belong in this range.

501 H

Start = B23900

End = B23900 + 0F10

= B24810H

∴ B123AB H does not belong in this range.

502 H

Start = B11C00

End = B11C00 + 0C12

= B12812 H

∴ B123AB H belongs in this range.

So, index number = 502 H

b) Start = B11C010 H

End = B12812 H

b) P.A. = B.A. + offset

= B11C00 + 2D4

= B11ED4 H

c & d) Access rights = D1 H

1101 0001

E = 0 (data/stack)

ED = 0

(Data Segment)

d) A = 1, So, the segment has been accessed.

e) DPL = 10

f) P = 1, So, the descripton is defined.

2. (a) $\frac{\text{Resolation of the ADC}}{\text{System Voltage}} = \frac{\text{ADC reading}}{\text{Analog voltage measured}}$

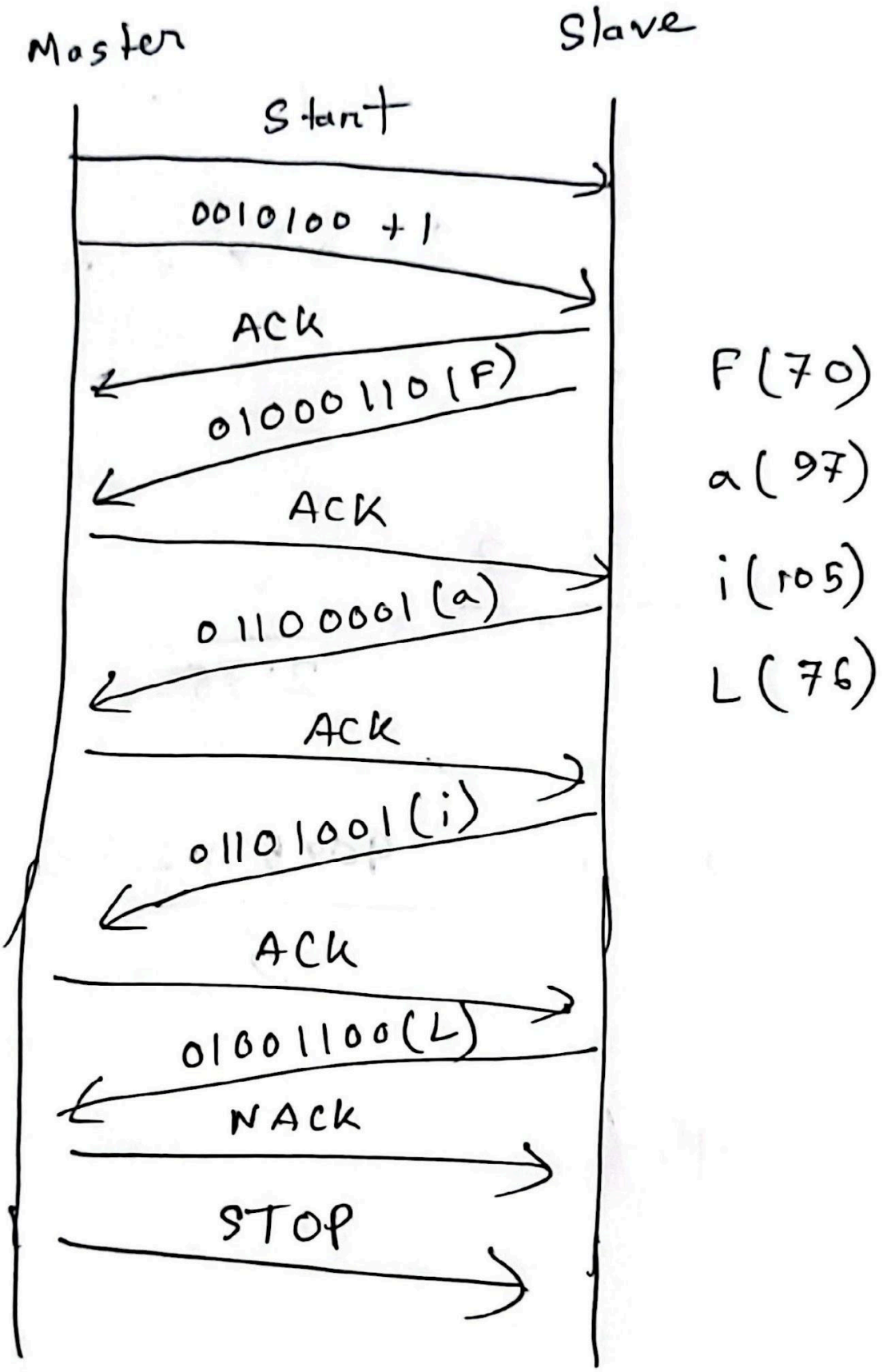
$\Rightarrow \frac{2^n - 1}{5} = \frac{2457}{3}$

$\Rightarrow 2^n - 1 = \frac{2457 \times 5}{3} = 4095$

$\Rightarrow 2^n = 4096 = 2^{12} \quad \therefore n = 12$

∴ To reduce quantization error, we need to increase the value of 'n' so that we can increase the no. of levels

2.(b)



F (70)

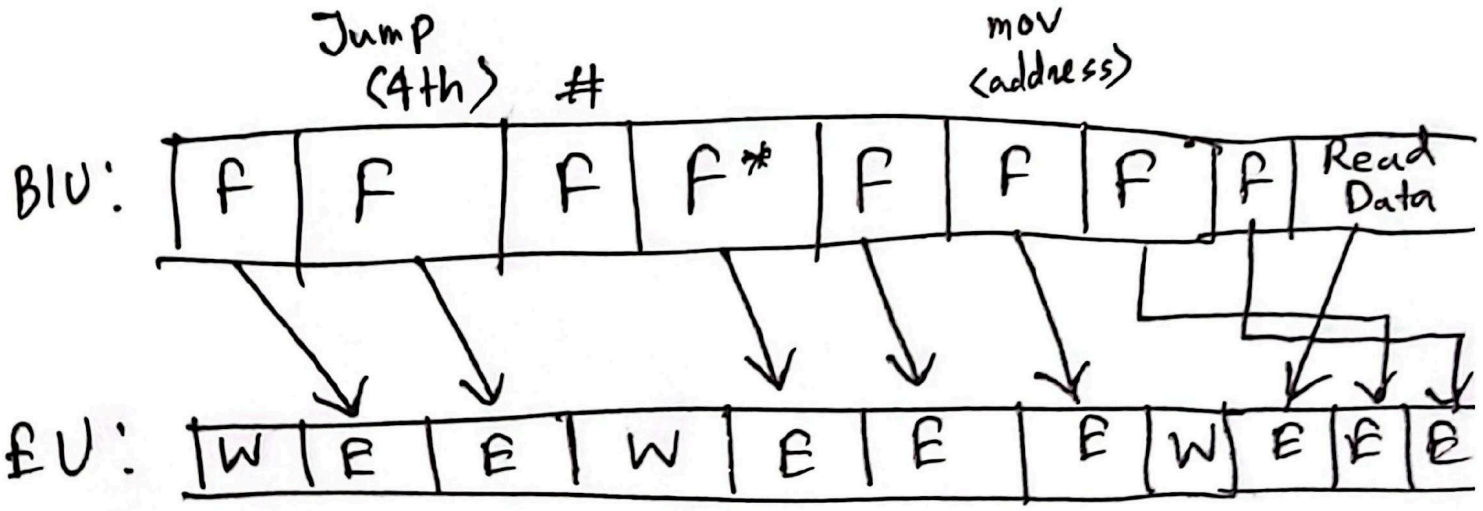
a (97)

i (105)

L (76)

c) Arduino would be more suitable.

3. (a)



(b) (I)

- The READY input causes wait states for slower memory and I/O components.
- A wait state is a situation in which a computer processor is waiting for the completion of some event before resuming activity. A program or process in a wait state is inactive for the duration of the wait state.
- When a computer processor works at a faster clock speed than the random access memory (RAM) that sends it instructions, it is set to go into a wait state for one or more clock cycles so that it is synchronized with RAM speed. In general, the more time a processor spends in wait states, the slower the performance of that processor.
- Wait states are a pure waste for a processor's performance. Modern designs try to eliminate or hide them using a variety of techniques: CPU caches, instruction pipelines, instruction prefetch , simultaneous multithreading and others.

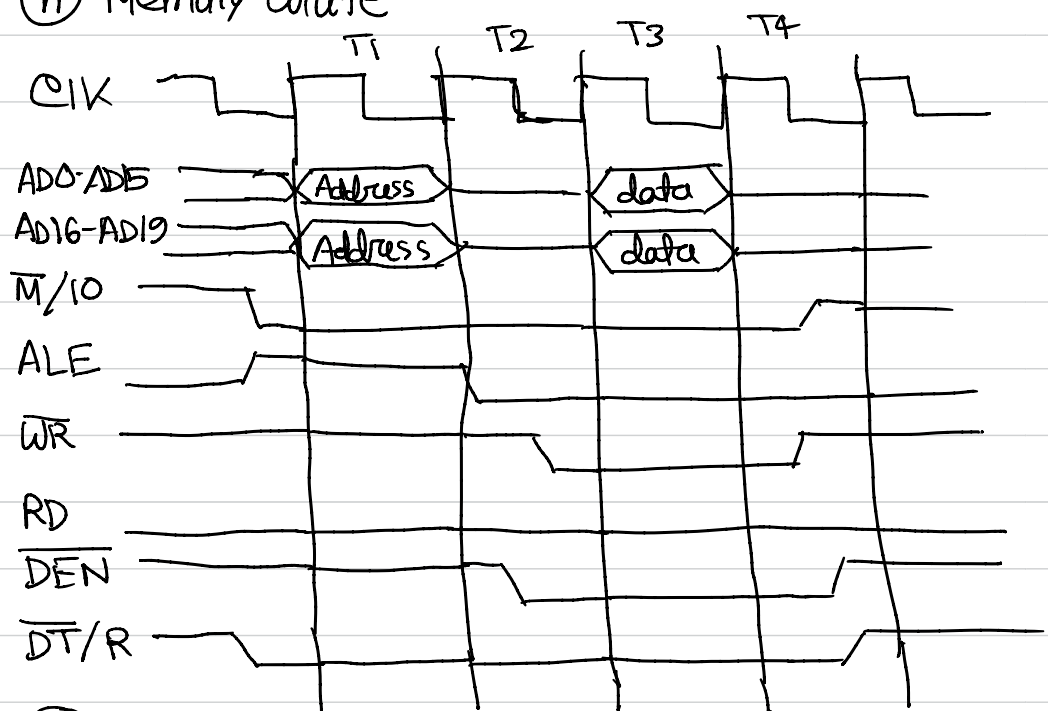
8

8086 Ready pin

The READY input is controlled to insert "Wait states" into the timing of the microprocessor. If the READY pin is placed at a logic 0 level, the micro-processor enters into wait states and remains idle. When it is high (logic 1), it indicates that the device is ready to transfer data.

Scanned with CamScanner

(n) Memory Write



(c)

Page fault

- If the page required by the processor is not in the main memory, then page fault occurs.
- The OS then reads the page from disk and puts in the RAM.
- The OS then changes the Page Table to map the new page.

Q4

a)

LDA FH
ADD FH
ADD FH
SUB CH
SUB CH
ADD DH
OUT
HLT

b)

Address	Content		Address	Content
0H	1001 1111		8H	XXXX XXXX
1H	0010 1111		9H	XXXX XXXX
2H	0010 1111		AH	XXXX XXXX
3H	0001 1100		BH	0000 1000
4H	0001 1100		CH	0000 0010
5H	0010 1101		DH	0000 0001
6H	0100 XXXX		EH	0000 0110
7H	0011 XXXX		FH	0000 0011

c) Control Word for **SUB CH**:

CON = $C_p E_p \overline{L_M}\,\overline{C_E}\;\; \overline{L_I}\,\overline{E_I}\,\overline{L_A} E_A\;\; S_U E_U \overline{L_B}\,\overline{L_O}$

T1: 5E3 (0101 1110 0011)

T2: BE3 (1011 1110 0011)

T3: 263 (0010 0110 0011)

—----------

T4: 1A3 (0001 1010 0011)

T5: 2E1 (0010 1110 0001)

T6: 3CF (0011 1100 1111)